

Vincent Do.

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EXPERIENCE

Boeing Company

Berkeley, MO

Software Engineer (Secret Security Clearance)

Aug 2024 – Present

- Shipped production C++ features delivering 6+ real-time navigation message streams to embedded display systems, backed by automated tests and Dockerized build workflows for end-to-end deployment
- Designed the message protocol and internal SDK for a real-time messaging framework adopted by engineers for integration
- Own integration and release readiness for a shared real-time framework used by 60+ dependent modules, resolving binary timing-drift issues before they surface as compatibility regressions
- Traced a long-standing cross-system defect to a binary data-merge edge case within two weeks of being introduced to a new system, restoring correct downstream output
- Led a month-long partner-software evaluation, integrating it with the messaging architecture to validate customer requirements at 1.5× required throughput and documenting release procedures

Expedia Group

Chicago, IL

Software Development Engineer Intern

May 2023 – Aug 2023

- Shipped customer-facing search features in the Expedia mobile app: VIP Access badges (A/B-tested, with lift on bookings and click-through), 10+ amenity filters, and clearer error handling that reduced booking-search friction
- Implemented GraphQL schema changes across 5+ Kotlin microservices, unblocking new mobile booking flows across backend and client surfaces and integrated backward-compatible iOS and Android production releases
- Built backend support for localized search across 16 languages, working with frontend and localization teams to validate dynamic rendering before release by creating mock GraphQL response

University of Wisconsin–Madison

Madison, WI

Biomedical Undergraduate Researcher

May 2021 – Aug 2021

- Analyzed 16,813 genes across 169 phenotypes using Python, R, graph traversal, and visualization, reducing the candidate set to 25 high-priority Zellweger genes through frequency thresholds, chromosomal-location filtering, and recursive traversal

PROJECTS

Playable Chess AI

[demo](#) | [code](#)

Python, PyTorch, CUDA, Flask, JavaScript

- Built a 7.2M-parameter hybrid ResNet + LSTM architecture, trained on 3.8M grandmaster positions with unlikelihood loss
- Cut per-epoch training time roughly 60x, from 20+ hours on CPU to about 20 minutes on GPU, by building a CUDA mixed-precision pipeline and a custom streaming dataset that avoids loading 3.8M positions into memory
- Deployed a playable full-stack version on Flask/Railway with a GitHub Pages frontend and in-browser Stockfish WASM, pairing neural-network move candidates with alpha-beta search across selectable opponents

Shortest Path Between Airports

[demo](#) | [code](#)

C++, React, WebGL

- Implemented BFS, Floyd-Warshall, and betweenness centrality in C++ over a 1,000+ airport graph, ranking the most influential nodes shown to uncover routing inefficiencies, validated by C++ unit tests across all graph operations

Exploding Chickens

[demo](#)

Node.js, MongoDB

- Built a real-time multiplayer Node.js backend with asynchronous game logic and persistent session states managed on MongoDB, keeping gameplay consistent across concurrent players over 1,600+ games

Volleyball Motion Tracker

[code](#)

Python, MediaPipe, YOLOv8, OpenCV, NumPy

- Built a per-frame volleyball spike analyzer fusing pose tracking and ball detection to extract 5+ kinematics

TECHNICAL SKILLS

Languages C++, Python, Java, C#, JavaScript, TypeScript, Kotlin, SQL

Frameworks React, Node.js, Flask, PyTorch, GraphQL, RESTful APIs, Cloudflare

Tools Docker, Jenkins, Git, Linux, GCP, PostgreSQL, MongoDB, Railway, GitHub Actions, CI/CD

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. Computer Science & Chemistry · GPA 3.6

Aug 2020 – May 2024